1. Cover Page

Geethanjali College of Engineering and Technology

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ADVANCED ENGINEERING PHYSICS LABORATORY WORKBOOK

B. Tech. I Year- I Semester

CSE (AI&ML), CSE (Cyber Security), CSE (Data Science) and EEE

2025-2026

DEPARTMENT OF FRESHMAN ENGINEERING

15. INDEX PAGE

Experiment No: **Date:**

1. V-I CHARACTERISTICS OF LIGHT EMITTING DIODE (LED)

Aim: To draw the V-I characteristics and to measure the Plank's constant using Light Emitting Diode (LED)

Apparatus: LED (680nm), voltmeter (0-5V), ammeter (0-10 mA) and DC regulated power supply (0-5V).

Formula:

$$E_g = h\vartheta = \frac{hc}{\lambda} = eV_0$$

$$h = \frac{eV_0\lambda}{c} \, J-S$$

Where ϑ- frequency of the emitted photons,
c- velocity of light, e-charge of electron,
λ- wavelength of light,
V_0-turn on voltage and
h-Plank's constant.

Circuit diagram:

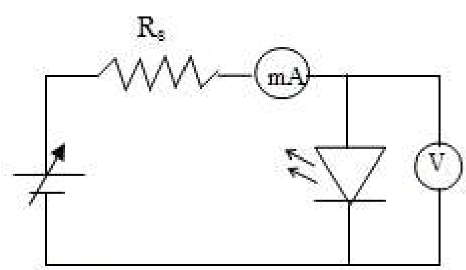


Fig.1(a): Circuit diagram to draw V-I Characteristics of LED

Procedure: -

1. Connect the LED in the forward bias as shown in circuit diagram.
2. Apply the forward bias voltage by using fine knob of regulated power supply.
3. Note the corresponding voltage and current values from the voltmeter and ammeter respectively.
4. See that initially the current drawn is negligible but it increases with the applied voltage. The LED starts glowing when it starts drawing the current.
5. Draw the V-I curve as shown in the model graph and find V_o (turn on voltage).

Theory:

In this experiment we will use light emitting diodes (LED's) to measure Planck's constant. LED's are semiconductors that emit electromagnetic radiation in optical and near optical frequencies when a voltage is applied to them. LED's emit light only when the voltage is forward biased and above a minimum threshold value. This combination of conditions creates an electron hole pair in a diode. Electron hole pairs are charge carriers and move when placed in an electrical potential. Thus, many electron hole pairs produce a current when placed in an electric field. Above the threshold value the current increases exponentially with voltage.

A quantum of energy is required to create an electron hole pair and this energy is released when an electron and a hole recombine. In most diodes this energy is absorbed by the semiconductor as heat, but in LED's this quantum of energy produces a photon of discreet energy $E_g = h\nu$. With increasing voltage, photons of increasing energies will be emitted. Thus, the light emitted by an LED may span a range of discrete wavelengths that decrease with increasing voltage above the threshold voltage (shorter wavelength = higher energy). We are interested in the maximum wavelength that is determined by the minimum energy needed to just to create an electron hole pair. It is numerically equal to the turn on voltage of the LED. The relation between the maximum wavelengths λ and the turn on voltage V_o is

$$E_g = h\vartheta = \frac{hc}{\lambda} = eV_0$$

Observations:

Wavelength = _________nm

S.No	Voltage (Volts)	Current(mA)

Model graph:

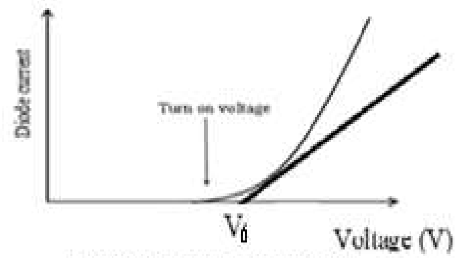


Fig.1(b): V-I characteristics of LED

Results:

1. Turn on voltage V_o = __________Volt.
2. The value of plank's constant determined from the experiment is ___________
3. The standard value of plank's constant is ___________
4. Percentage of error in the result obtained.
5. If any deviation observed in the result obtained vis-à-vis the standard result, specify the reason and justify it.
6. What are the final inferences and conclusions?

VIVA VOCE

1. **What are direct band gap semiconductors.**

2. **Mention any two applications of direct band gap semiconductors.**

3. **What are indirect band gap semiconductors?**

4. **Give an example for direct and indirect band gap semiconductors.**

5. **What is the principal of LED?**

6. **What is Turn on voltage?**

7. **Mention any two applications of LED.**

Experiment No: **Date:**

2. ENERGY GAP OF A SEMICONDUCTOR

Aim: To determine the energy gap of a given semiconductor.

Apparatus: Semiconductor diode, micro ammeter, thermometer, copper vessel, regulated DC power supply, heater and Bakelite lid.

Formula: $E_g = \frac{2x\ 2.303\ x\ \text{Slope}\ x\ k}{1.6\ x\ 10^{-19}}\ eV$ __________ (1)

Where E_g = energy gap of semiconductor,

k = Boltzmann's constant= 1.38 x 10^{-23}J/K

Slope = $\frac{dy}{dx}$ from graph between $\frac{1}{T}$ and $log_{10}I_s$, and

1 Joule = 1.6 x 10^{-19}eV.

Theory:

In a reverse biased pn junction diode, the reverse saturation current in the diode is independent of the applied voltage and is temperature dependent. Under appropriate experimental conditions the current flowing through a reverse bias pn junction diode is given by the equation,

$$I = A\left[e^{\left(-\frac{E_g}{kT}+\frac{eV}{kT}\right)}\right]$$

Where A - constant for a particular semiconductor taken,

E_g- energy gap of semiconductor,

k- Boltzmann constant,

T- Absolute temperature,

e- charge of the electron,

V- voltage across the p-n junction diode.

A graph can be drawn by taking $\frac{1}{T}$ on X-axis and the current passing through the semiconductor $\log_{10}I_s$ on Y-axis and by using the value of the slope; energy gap E_g can be computed using equation 1.

Description:

The apparatus consists of a diode, attached to a Bakelite lid with terminals. The diode can be immersed in oil, taken in the copper vessel. The Bakelite lid is provided with a hole to insert a thermometer. With the help of a heater, the oil can be heated to a desired temperature.

Procedure:

1. Make the circuit connections as shown in the Fig.2(a).
2. Pour some oil in the copper vessel, and fix the Bakelite lid on the copper vessel.
3. Insert thermometer into the vessel and apply a constant voltage (1.5V) to the given semiconductor. Increase the temperature of the oil to 80°C by switching on the heater.
4. Switch off the heater then the oil starts cooling.

5. Take the Current readings for every 5°C decrease of temperature from 80°C to 40°C, and note down in the table.

Circuit diagram:

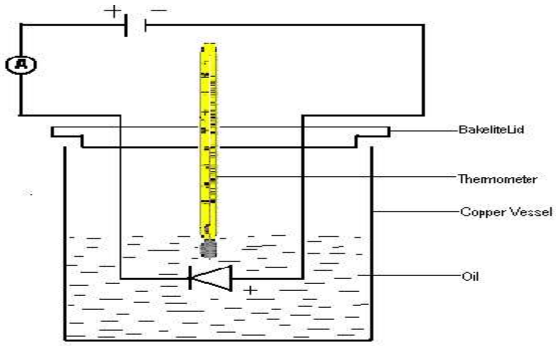


Fig.2(a): Circuit diagram to find E_g of semiconductor

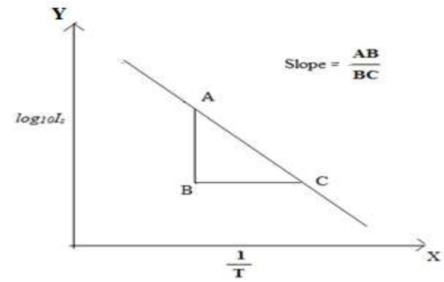


Fig.2(b): Graph between $\frac{1}{T}$ and $\log I_s$

Graph:

A graph is plotted by taking $log_{10}I_s$ on Y-axis and $\frac{1}{T}$ on X-axis.
A straight line is obtained as shown, whose slope is determined.

Observations:

Applied constant voltage (V) = 1.5 V.

S.No	Temperature		Current (I) µA	Log_{10} Is	1/T K^{-1}
	℃	K			
1	80				
2	75				
3	70				
4	65				
5	60				
6	55				
7	50				
8	45				
9	40				
10	35				

Calculations:

$$E_g = \frac{2\text{x } 2.303 \text{ x Slope x k}}{1.6 \text{ x } 10^{-19}}\ eV \qquad (1 \text{ Joule} = 1.6 \text{ x } 10^{-19}\ eV)$$

Substituting the value of slope in the formula, the energy gap is calculated.

$$E_g =$$

Results:

1. Expected value of Energy gap of the given semiconductor diode is ____________ eV.
2. Energy gap of the given semiconductor diode is obtained through experiment (graph) is ____________ eV.
3. Percentage of error in the result obtained.
4. If any deviation observed in the result obtained vis-à-vis the standard result, specify the reason and justify it.

Precautions:

1. The oil should not be heated above the stipulated temperature i.e. around 80°C.
2. The micro ammeter reading should not go beyond the scale limits.

VIVA VOCE

1. What is valence band energy?

2. Define conduction band energy.

3. What is energy band gap?

4. Explain intrinsic and extrinsic semiconductors.

5. Compare p-type and n-type semiconductors.

6. What is the importance of energy gap in a semiconductor?

7. Explain why resistance decreases with increase in temperature in a semiconductor.

Experiment No: **Date:**

3. V-I CHARECTERISTICS OF SOLAR CELL

Aim: To study the V-I characteristics of Solar Cell.

Apparatus: Voltmeter, ammeter, variable load, solar cell and mercury coated variable intensity source.

Formula:

Fill Factor $(FF) = \frac{P_{max}}{V_{OC}I_{SC}}$

Where P_{max}Is the maximum power from graph,

V_{OC} is the open circuit voltage,

I_{SC} is the short circuit current.

Description: It is a p-n junction device with a large surface area based on the principle of Photovoltaic effect. It directly converts light energy into electrical energy and hence is also known as Photovoltaic cell. In solar cell almost the total radiation (light) incident is absorbed and electron-hole pairs are generated giving raise to electrical energy. It has high efficiency. The materials used for solar cells are Si, CdS, GaAs etc. The solar cells are broadly classified into a) p-n homojunction type, b) p-n hetero junction type, c) Metal-Insulator-Semiconductor (MIS), and d) Semiconductor-Insulator-Semiconductor (SIS).

Efficiency of solar cell is defined as the ratio of the total power converted by the solar cell to the total power available for energy conversion.

Procedure:

1) Connect the circuit as shown in the circuit diagram.
2) Place Solar Cell directly in front of variable light intensity source and connect output of Solar Cell to voltmeter.
3) For various intensities of light (bulb) note down voltmeter and ammeter readings.
4) The experiment is repeated with a variable load in the circuit as shown.
5) Plot a graph, between voltage and current at different intensities with & without load.

Circuit diagram:

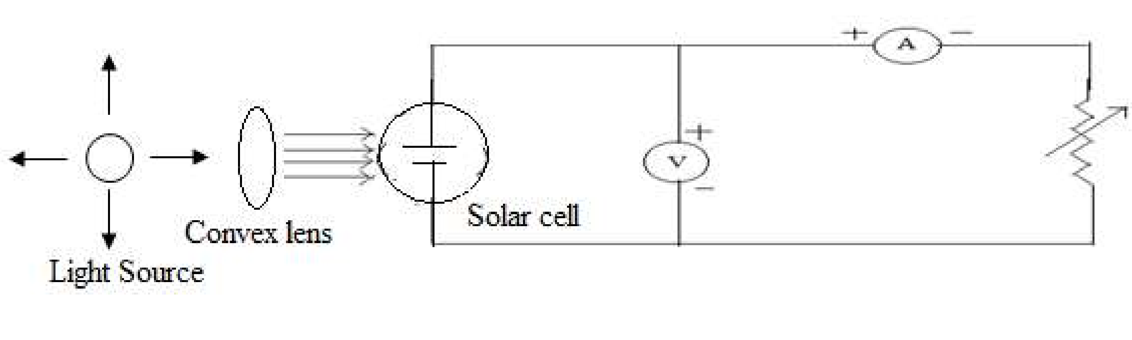


Fig. 3(a): circuit diagram to plot V-I characteristics of solar cell

Observations:

S.No.	Without load	
	Voltage (V)	Current (I) (mA)
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

S.No.	With load		
	Voltage (V)	Current (I) (mA)	Power(P)(Watt)
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

Model graph:

V-I Characteristic curve without load

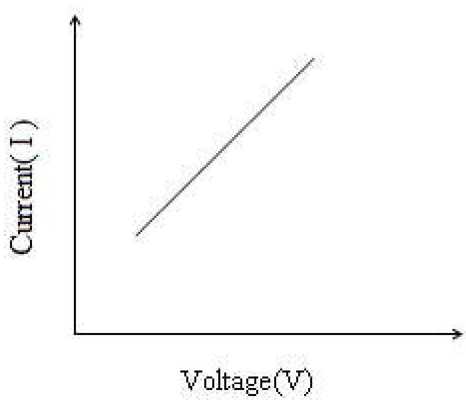


Fig.3(b) : V-I Characteristics of solar cell without load

V-I Characteristic curve with load

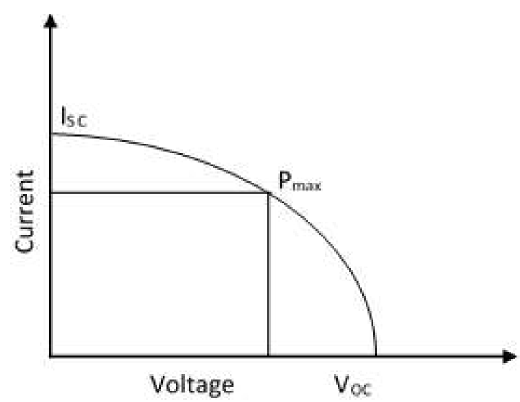


Fig.3(c) : V-I Characteristics of solar cell with load

Results:

1. V- I characteristic curve of a given solar cell obtained through experiment (Graph).
2. Is there any deviation in V-I characteristic curve of a given solar cell?
3. If any deviation observed in the result obtained vis-à-vis the expected/standard result, specify the reason and justify it.
4. What are the final inferences and conclusions?

VIVA VOCE

1. **What is solar cell?**

2. **Which type of materials are used in solar cells?**

3. **What is the requirement of a solar cell material?**

4. **What is short circuit current?**

5. **What is open circuit voltage?**

6. **What is efficiency of a solar cell?**

7. **What is the principle of solar cell?**

8. **What is the difference between photodiode and solar cell?**

9. **Write any two applications of solar cell.**

Experiment No: **Date:**

4. HALL EFFECT

Aim: **T**o determine the

i) Hall coefficient of a semiconductor
ii) type of the semiconductor
iii) concentration of the majority carriers and
iv) mobility of the majority carriers in the material.

Apparatus: An extrinsic semiconductor wafer, a constant current source, an electromagnet anda voltmeter having high input impedance.

Formulae:

i) The sign of the majority carriers is found from the polarity of Hall voltage V_H

ii) The Hall coefficient R_H is calculated from,

$$R_H = \frac{V_H t}{BI}$$

iii) The concentration of the majority charge carriers is

$$p = \frac{1}{R_H e} \qquad \text{or } n = \frac{1}{R_H e}$$

iv) The mobility of the charge carriers is

$$\mu = \sigma R_H$$

Circuit diagram:

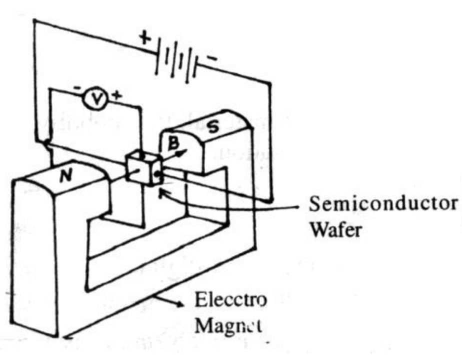


Fig.4(a) : Basic experimental arrangement to study Hall effect

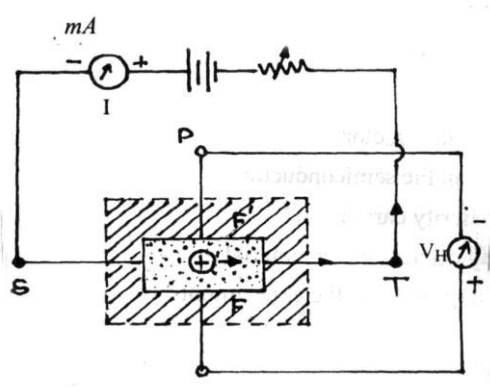


Fig.4(b) : Illustration of Measurement of Hall voltage

Procedure:

1. The semiconductor sample is carefully mounted on the four probe strip (Hall sensor or Hall probe) and electrical contacts are provided. Two probes are arranged for current supply to sample (input) and other two probes arranged for measuring voltage (output). These four terminals are connected to the Hall Effect set up.
3. The lengthwise contacts of the probe are connected to the current terminals of the set up. The widthwise contacts of the Hall probe are connected to the voltage terminals of the Hall Effect set up.
4. The probe should be in a central position in the air gap of the electromagnet set up. The power supply of electromagnetic setup is switched on, **gradually increase the power supply** and the magnetic field is adjusted to a suitable value (say 3 kilogauss).
5. Hall Effect set up is switched on and the Hall probe is adjusted in a vertical plane and center of electromagnetic setup till the Hall voltage generated is a maximum.
6. The current is varied and the corresponding Hall voltage is noted down. The observations are recorded in observation table. The Hall voltage and its polarity are noted down. After the results, gradually decrease the power supply of electromagnetic setup to zero.
7. Other way to measure the hall voltage is, the current is kept at a constant value, say 3mA. The magnetic field is varied in step size of 500 gauss. At each step of the magnetic field, the corresponding Hall voltage is noted down. The observations are recorded in observation table. The Hall voltage and its polarity are noted down.
8. Using the above observations, graphs are plotted. In one graph, the Hall voltage is plotted as a function of current at a constant magnetic field. In a second, the Hall voltage is plotted as a function of magnetic field at constant current. In both the graphs straight line plots are obtained, as shown in fig.10(c) and fig. 10(d). The slope of straight line is determined in each case.
9. Using the values of slopes into equation the value of R_H in each case is calculated. The mean of two values is taken as the value of R_H.
10. Substituting the value of R_H into equation the concentration of charge carriers is determined.
11. From the manufacturer's data, the resistivity of the semiconductor sample is given and using the value in equation, the carrier mobility is to be evaluated.

Observations:

Thickness of the semiconductor wafer (t): ________________ mm. = ___________________ m.

Resistivity (ρ): _________________________ ohm. m.

Magnetic field, B = ____________k gauss = _____________Wb/m^2 (1 wb/m^2 = 10^4 gauss)

S. No.	Current I (mA)	Hall voltage V_H (mV)
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

Current (I):_____________mA =_____________ x 10^{-3} A

S. No.	Magnetic field		Hall voltage V_H (mV)
	K gauss	Wb/m^2	
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

Calculations:

$$\text{Slope } \frac{V_H}{I} = \frac{PR}{QR} =$$

$$\text{Slope } \frac{V_H}{B} = \frac{PR}{QR} =$$

i) R_{H1} (Slope $\frac{V_H}{I}$) x $\frac{t}{B}$ = ______________ = ____________ m^3/C

R_{H2} (Slope $\frac{V_H}{B}$) x $\frac{t}{I}$ = ______________ = ____________ m^3/C

Mean $R_H = \frac{RH_1 + RH_2}{2}$ = ________________________ m^3/C

ii) Majority carrier concentration

$$n \text{ (or p)} = \frac{1}{R_H e} = \frac{1}{............m^3/C x 1.602 \times 10^{-19} C}$$

= ____________________ / m^3

iii) Mobility of the charge carriers

$\mu = \sigma R_H$ = ____________ S/m x_____________ m^3/C

= ___________________ $m^2/V.s$

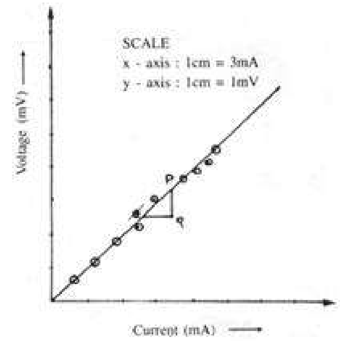


Fig. 4(c): Variation of Hall voltage with current

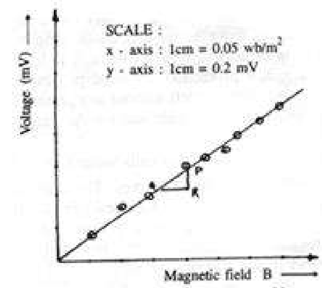


Fig. 4(d): Variation of Hall voltage with magnetic field

Results:

The polarity of the Hall voltage is __________________.

Therefore, the given semiconductor is of _______ type.

The value of Hall coefficient is ________________ m^3/C.

The concentration of the majority carriers is found to be _______________ $/m^3$.

The mobility of the charge carriers is found to be ___________________$m^2/V.s$.

Precautions:

1. Care is taken to limit the current through the probe to a value less than mentioned by the manufacturer.
2. The probe is properly centered and oriented in the magnetic field such that maximum Hall voltage is generated.
3. Magnetic field is varied gradually in steps to avoid damage to the electromagnet coils.

VIVA VOCE

1. **What parameters can you determine with this experiment?**

2. **State and explain Hall Effect.**

3. **How do you determine mobility of a charge carrier using Hall effect?**

4. **What is Hall coefficient?**

5. **How does mobility depends on electrical conductivity?**

6) What are the applications of Hall effect?

Experiment No: **Date:**

5. NUMERICAL APERTURE (N.A) & BENDING LOSSES OF A GIVEN OPTICAL FIBER

Aim: To determine the numerical aperture, acceptance angle and bending losses of a given optical fiber.

Apparatus: Optical fiber cable (1,3,5 meter), Digital Multimeter (DMM), fiber optic trainer module kit.

Formula: $\text{Numerical Aperture (N.A.)} = \frac{W}{(4L^2+W^2)^{1/2}}$ -------- (1)

Here, W is the Diameter of the light spot falling on the screen.

L is the distance between fiber end and screen.

Acceptance angle (θ_a) = $\sin^{-1}$(N A) ----------------- (2)

Loss of optical power = $-10\log(P_{out}/P_{in})$dB -----------(3)

Most fiber manufactures characterize attenuation loss by the number of decibels loss per kilometer of fiber. This value can be calculated by

Loss/km = $-(10/L)\log(P_{out}/P_{in})$dB/km ---------------(4)

Description: The fiber optic trainer module consists of three important parts namely,

1. Electrical to optical (E/O) converter
2. Optical to electrical (O/E) converter
3. Optical power meter

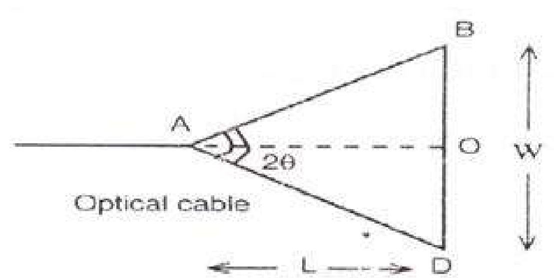
Theory: The numerical aperture of an optical fiber is a measure of the light gathering capacity of the optical fiber. It is given by

Numerical aperture (N.A) = $n\sin\theta_a$

Where n is the refractive index of the medium.

For air n = 1

$\therefore \text{N.A.} = \sin\theta_a$

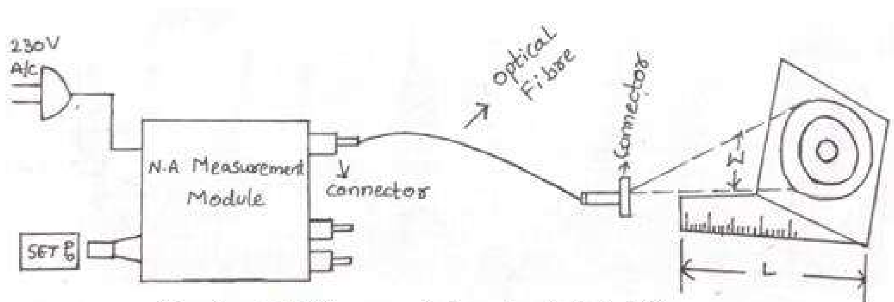


Geomatrical method of measuring numerical aperture

For step-index fiber, the N.A. is given by:

$$\mathrm{N.A.} = (n_{\mathrm{core}}^2 - n_{\mathrm{cladding}}^2)^{1/2}$$

Light from the fiber end "A" is incident on the screen BD. Let the diameter of the light spot falling on the screen=BD



Experimental kit for numerical aperture in Optical fiber

Let the distance between the fiber end and the screen=AO=L.

$\therefore$ From the fig.(a), $\mathrm{AB} = \left(\mathrm{L}^2 + \frac{\mathrm{W}^2}{4}\right)^{1/2} = \frac{(4\mathrm{L}^2+\mathrm{W}^2)^{1/2}}{2}$

$\therefore$ $\mathrm{N.A} = \sin\theta_a = \frac{\mathrm{OB}}{\mathrm{AB}} = \frac{\mathrm{W}}{(4\mathrm{L}^2+\mathrm{W}^2)^{1/2}}$ Knowing W and L, the N.A can be calculated by using equation (1). Substituting this value of N.A. in equation (2), the acceptance angle θ_a can be calculated.

(a) Procedure to determine the numerical aperture:

1. Connect one end of the fiber optic cable to $P_0$and the other end to N.A.jig as shown in figure.
2. The AC main is switched ON. Light should appear at the end of the fiber on the N.A. jig to ensure proper coupling is made or not. Turn the SET P_0 knob in clockwise direction to get maximum intensity light through fiber.
3. A screen with concentric circles of known diameter is kept vertically at a distance of (L) from the fiber end and view red spot on the screen.
4. The diameter (W) of the red spot is made exactly equal to the first concentric circle and the corresponding distances (L) from the fiber end to the screen are noted in table 1.
5. The experiment is repeated for the subsequent diameters of the circles by adjusting the length L.

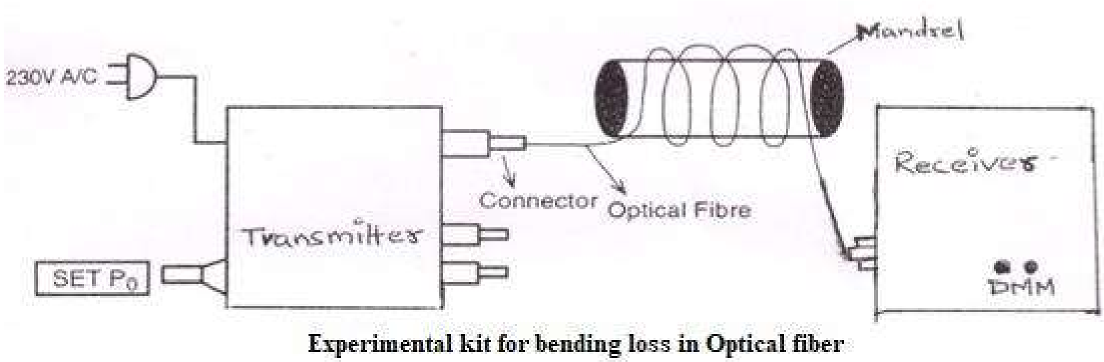
Table 1:

S.No.	W (mm)	L (mm)	$(N.A.) = \dfrac{W}{(4L^2 + W^2)^{1/2}}$	$\theta_a = \sin^{-1}(N.A)$
1	10			
2	20			
3	30			
4	40			
5	50			

(b) Procedure to determine bending losses in Optical Fiber:

In this part of the experiment a transmitter and receiver along with mandrel are used.

1. One end of the optical fiber is connected to transmitter and other end is connected the receiver.
2. The knob P_0 of the transmitter kit is adjusted to get the maximum intensity.
3. The receiver reading is noted with the help of a digital multimeter when there are no bends in the fiber, the reading is recorded as P_{in}.
4. The optical fiber is wound on the mandrel say by 5 turns and the receiver reading is noted as P_{out} in digital multimeter and recorded in the table 2.
5. The bending loss is calculated the above formula.
6. The experiment is repeated for more no. of turn's say 10, 15….turns and the corresponding readings are tabulated.



Experimental kit for bending loss in Optical fiber

Table 2:

Digital multimeter reading without bending of the optical fiber P_{in}= ____________ mW

S.No	Length of optical fiber-L(m)	No. of turns on the mandrel	P_{out} µW	Bending loss dB/Km [$-(10/L) \log (P_{out}/P_{in})$dB/km]
1	1			
2				
3				
4	3			
5				
6				
7	5			
8				
9				

Precautions:

1. The knob P_0 should be adjusted for the maximum optical output.
2. There should not be any bends in optical fiber while noting down P_{01}.

Result:

1. The Numerical aperture of the optical fiber =______________________.
2. The acceptance angle of the optical fiber = ___________________.
3. The bending losses in optical fiber for various turns are determined.

Applications:

Optical fibers find applications in

1. Sensors
2. In the field of medicine (Endoscopy).
3. Military applications (In submarines, missiles and air craft's)
4. In the field of Communication systems.

VIVA VOCE:

1. **What is the Principle of Optical Fiber?**
2. **What is Numerical Aperture?**
3. **What is acceptance angle?**
4. **What is meant by mono mode and multimode fibers?**
5. **What is attenuation in Optical fiber?**
6. **What are the various types of attenuation in Optical fiber?**
7. **What is the source of light used in this experiment?**
8. **Intermodel dispersion is negligible in what type of Optical fiber?**
9. **What are the advantages of optical fiber?**
10. **What are the applications of optical fiber?**

Experiment No: **Date:**

6. Dielectric Constant

Aim: To determine the dielectric constant of the dielectric medium present in a parallel plate capacitor.

Apparatus: DC regulated power supply, resistors, electrolytic capacitors (capacitor with large value), digital voltmeter, stop clock, connecting wires.

Theory: An electrolytic capacitor (e-cap) is a polarized capacitor whose anode (positive plate) is made of a metal that forms an insulating oxide layer through anodization. This oxide layer acts as the dielectric of the capacitor. A solid, liquid, or gel electrolyte covers the surface of this oxide layer, serving as the (cathode) or negative plate of the capacitor. Due to their very thin dielectric oxide layer and enlarged anode surface, electrolytic capacitors have a much higher capacitance-voltage (CV) product per unit volume compared to ceramic capacitors or film capacitors, and so can have large capacitance values.

There are three families of electrolytic capacitor: aluminum electrolytic capacitors, tantalum electrolytic capacitors, and niobium electrolytic capacitors. The large capacitance of electrolytic capacitors makes them particularly suitable for passing or bypassing low-frequency signals, and for storing large amounts of energy. They are widely used for decoupling or noise filtering in power supplies and DC link circuits for variable-frequency drives, for coupling signals between amplifier stages, and storing energy as in a flashlamp.

Electrolytic capacitors are polarized components due to their asymmetrical construction, and must be operated with a higher voltage (ie, more positive) on the anode than on the cathode at all times. For this reason, the anode terminal is marked with a plus sign and the cathode with a minus sign. Applying a reverse polarity voltage, or a voltage exceeding the maximum rated working voltage of as little as 1 or 1.5 volts, can destroy the dielectric and thus the capacitor. The failure of electrolytic capacitors can be hazardous, resulting in an explosion or fire. Bipolar electrolytic (aka non-polarized) capacitors which may be operated with either polarity are special constructions with two anodes connected in series.

Formula: The dielectric constant of the material in the given capacitor can be calculated by an equation

$$K = \frac{d \; X \; T_{\frac{1}{2}} \; X \; 10^{-6}}{0.693 \; X \; (\varepsilon_0 A R)}$$

Where: 'K' - dielectric constant, 'd' – thickness of the dielectric material

'$T_{\frac{1}{2}}$' - time required to charge or discharge a capacitor to 50%.

'ε_0' - is the permittivity of free space, 'A'- area of the dielectric material.

'R' – resistor connected in the circuit

Charging:

Capacitors are devices which store electric energy by means of an electrostatic field and release this energy later. The voltage across the capacitor when it gets charged gradually at any instant of time t, is given by

Applying Kirchhoff voltage law across the RC circuit (Fig.1) and solving, we get,

$$i(\mathrm{t}) = \frac{V_0}{R} e^{\frac{-t}{RC}} \quad \text{---------------} \quad (1)$$

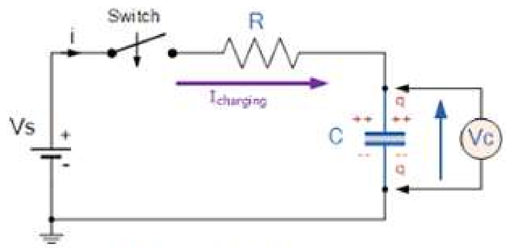


Fig. 1: Charging of the Capacitor

$$V_c(t) = V_0 - V_R(t) = V_0 - R.\ i(t)$$

$$= V_0\left(1 - e^{\frac{-t}{RC}}\right) \quad \text{-----------} \quad (2)$$

Where V_0 - maximum voltage when the capacitor is fully charged.

The equation (2) gives the variation of voltage across the capacitor with the time.

At time t = RC, $V_c(t) = V_0\left\{1-\left(\frac{1}{e}\right)\right\} = 0.63\ V_0$, i.e. time constant is equal to the time taken to increase the voltage across the capacitor to 0.63 times of the maximum voltage.

Discharging:

The voltage across the capacitor, during the discharging phase, $V_c(t) = V_0\left(e^{-\frac{t}{RC}}\right)$

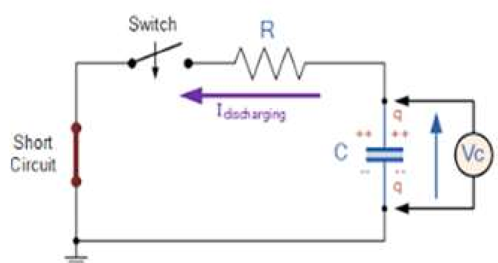


Fig.2: Discharging of the Capacitor

At time t = RC, $V_c(t) = V_0\frac{1}{e} = 0.37\ V_0$ i.e. time constant t is equal to the time taken to decrease the voltage across the capacitor to 0.37 times of the maximum voltage.

Let $T_{\frac{1}{2}}$ be the time required to charge or discharge a capacitor to 50%.

i.e. when t = $T_{\frac{1}{2}}$, $V_c = \frac{V}{2} = Ve^{\frac{-t}{RC}}$, taking logarithm on both sides $e^{\frac{t}{RC}}$ = ln2 =0.63

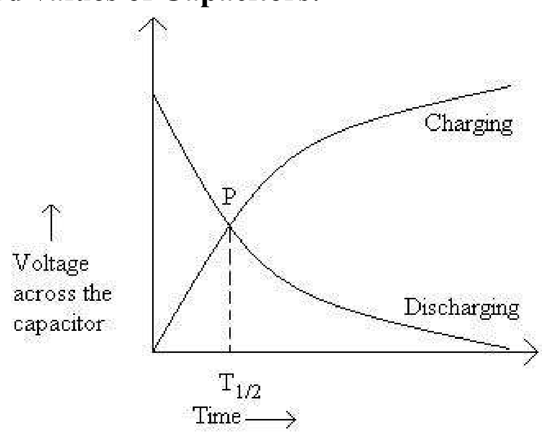
$C = \frac{T_{\frac{1}{2}}}{0.63\ R}$ but the capacity of the capacitor $C = \frac{K\ \varepsilon_0 A}{d}$, where 'A' and 'd' are the thickness and area of the dielectric material. 'ε_0' is the permittivity of free space and 'K' dielectric constant,

$$K = \frac{d \, X \, T_{\frac{1}{2}} \, X \, 10^{-6}}{0.693 \, X \, (\varepsilon_0 AR)}$$

Procedure:

1. Make the connections as shown in Fig.1.
2. Note the values of the resistor and capacitor and hence determine the theoretical time constant of the circuit RC.
3. Switch on the regulated power supply and adjust it to a voltage of 5V.
4. Close the switch and start the stop clock simultaneously.
5. Take the readings of the capacitor voltage for every 5 sec. Continue this until to get the saturation voltage (maximum value of voltage).
6. Replace regulated power supply by a short circuit as shown in Fig. 3.
7. Close the switch and restart the stop clock simultaneously.
8. Take readings of capacitor voltage for every 5 sec until the capacitor is completely discharged.
9. Plot the charging and discharging curves (V versus t). The charging mode curve and the discharging mode curve intersect at the point P.
10. By referring the position 'P' to the time axis, the value of its abscissa $T_{\frac{1}{2}}$ in seconds is found out and the dielectric constant 'K' can be calculated using the formula.

Model graph and Standard values of Capacitors:



Capacitance Value	Length (L) mm	Breadth (b) mm	Thickness (d) mm
3300 µF	50	7	0.1
2200 µF	40	6	0.1
1000 µF	40	5	0.1
470 µF	30	5	0.1

Observation table:

Time(in seconds)	R =_______Ω,	C = ________µF
	Charging (V)	*Discharging(V)*
0		
5		
10		
15		
20		
25		
30		
35		
40		
45		
50		
55		
60		
65		
70		
75		
80		
85		
90		
95		
100		
105		
110		
115		
120		
125		
130		
135		
140		
145		
150		
155		
160		

CALCULATIONS:

Resistor R =
Length L =
Breadth b =
Thickness d =
Area L x b =
Time (S) =

$$K = \frac{d\ X\ T_{\frac{1}{2}}\ X\ 10^{-6}}{0.693\ X\ (\varepsilon_0 AR)}$$

Results:

1. Standard value of dielectric constant of the material in the given capacitor = _________.
2. Dielectric constant of the material in the given capacitor obtained through experiment _________.
3. Percentage of error in the result obtained.
4. If any deviation observed in the result obtained vis-à-vis the expected/standard result, specify the reason and justify it.

Precautions:

1. Starting the stop clock and noting the deflection in the voltmeter should be simultaneous.
2. The voltmeter reading should not go beyond the scale limits.

VIVA VOCE

1. **Define dielectric constant.**

2. **Explain charging and discharging in RC circuit?**

3. **What are the units for dielectric constant?**

4. **Which material has high dielectric constant?**

5. **How time constant is related to charge and discharge processes?**

6. **What happens to the capacitor when the capacitor voltage is equal to the source voltage?**

7. **Define permittivity.**

8. **Explain different types of dielectrics.**

9. **Mention few applications of RC circuit.**

Experiment No: **Date:**

7. Determination of magnetic moment of a bar magnet and horizontal earth magnetic field

AIM: To determine the magnetic moment M of a bar magnet and to find the absolute value H of horizontal component of earth's magnetic field by using deflection and vibration magnetometers.

APPARATUS: A bar magnet whose M is to be determined, a deflection magnetometer, a vibration magnetometer, a brass bar of about the same mass and size as the magnet, a compass needle, a stop watch, a Vernier calliperse, meter scale and a balance.

PRINCIPLE & FORMULA:

To determine the values of M and H, the experiment is to be performed in two parts.

PART I: A deflection experiment in which the bar magnet is used to deflect a magnetic needle in a tangent position of Gauss. If θ be the deflection of the needle from the magnetic meridian when the center of the bar magnet is at a distance 'd' from it, we know that for tan A position

$$\frac{M}{H} = \frac{(d^2 - l^2)^2}{2d} tan\theta \qquad (1)$$

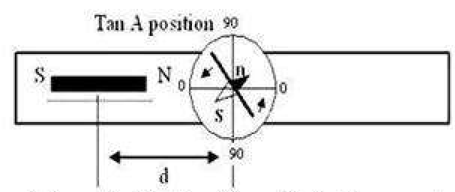


Figure 1 shows the Tan A position of deflection magnetometer.

where M = Magnetic moment of the magnet

l = Half the distance between its poles

H = Horizontal component of earth's magnetic field.

PART-II: An oscillation experiment in which the same bar magnet is allowed to oscillate in earth's field alone. If T be the period of oscillation of the magnet, then from the expression for the time period of simple harmonic motion.

We have $$T = 2\pi\sqrt{\frac{I}{MH}}$$

$$MH = 4\pi^2 \frac{I}{T^2} \qquad (2)$$

where I = movement of inertia of the magnet about the vertical axis through its center of gravity.

For rectangular magnet $$I = \frac{m(a^2+b^2)}{12} \qquad (3)$$

Where m = mass of the magnet, a = length of the magnet, b = breadth of the magnet multiplying equation (1) and (2) and taking the square root, we have

$$M = \frac{\pi(d^2 - l^2)}{T}\sqrt{\frac{2I\,Tan\theta}{d}} \qquad (4)$$

which gives the value of M in web/cm

Dividing equation (2) by equation (1) and taking the square root, we have

$$H = \frac{2\pi}{T(d^2 - l^2)}\sqrt{\frac{2Id}{Tan\theta}} \qquad (5)$$

which gives the value of H in Oersted.

PROCEDURE:

I. DEFLECTION EXPERIMENT

1. Place the deflection magnetometer on a rigid horizontal table and turn it until arms are parallel to the aluminum pointer.
2. Rotate the compass box until the aluminum pointer reads O-O on the circular scale keeping the arms of the magnetometer parallel to the pointer. This sets the magnetometer for tan A position (Fig.1).
3. Place the magnet whose magnetic movement is to be determined on one of the arms of the magnetometer, then after gently tapping the compass box read the positions of both ends of the pointer on the circular scale. Note down the distance 'd' from the center of the magnet to the center of the needle.
4. Then reverse the magnet so that its north and south poles are interchanged. Keeping its distance same and again note down the deflections at the two ends of the pointer.
5. Transfer the magnet to the second arm of the magnetometer and keeping its distance as same, repeat the above determinations.
6. Take down the mean of the 8 values of deflection of the needle which gives the mean value of the θ.
7. Repeat the above observations for different values of 'd'.

II. OSCILLATION EXPERIMENT:

1. Represent the magnetic meridian on rigid horizontal table by a light chalk line, drawn with the help of a compass needle and a meter scale.
2. Place the vibration magnetometer on the table with its longer side parallel to the magnetic north and south line.
3. Place the compass needle inside up on the line mark on the plane mirror fixed the bass of the box and adjust the magnetometer.
4. Place in the stirrup a brass bar of about same mass and size as the magnet. When the bar is at rest observe the angle between the bar and the index line on the plane mirror beneath it and turn the torsion bead through this angle so that the brass bar may rest parallel to the index line. Now holding the stirrup tight in position so that no further twisting of the fibre takes place, remove the brass bar and put in the magnet used in the deflection experiment with its N-pole pointing towards magnetic north.
5. Now rotate the suspended magnet through a small angle about it axis of suspension by bringing slowly and cautiously towards it another magnet presented end on with its north pole towards the north pole of the suspended magnet. Take away the second magnet immediately to a sufficient distance from the suspended magnet when the later will start oscillating about its position of rest.
6. Determine time period twice with an accurate stop watch.
7. Repeat the observations for period of magnet with altered number of oscillations and find out the mean value of T.

OBSERVATIONS:

Mass of the magnet (m) = ________ gm, Length of the magnet ($2l$) = _______ cm

Breadth of the magnet (b) = _______ cm, Mean time period of oscillation (T) = ______ Sec.

Distance between center of the magnet and needle (d) = _____ cm

Mean Tan θ = _______

PRECAUTIONS:

1. The magnetometer should be placed on a rigid table. The base of the magnetometers must be in perfect horizontal level and once the instrument have been adjusted they should not be disturbed throughout the whole experiment.

2. All pieces of magnetic materials and current bearing conductors should be kept at a considerable distance from the magnetometers.

Table-I

As, $M = \frac{\pi(d^2-l^2)}{T}\sqrt{\frac{2I\,Tan\theta}{d}}$ weber/cm, $H = \frac{2\pi}{T(d^2-l^2)}\sqrt{\frac{2Id}{Tan\theta}}$ Oersted and $I = \frac{m(a^2+b^2)}{12}$ gm-cm^2

Determination of Tanθ

S.No	Distance cm	Deflection of the pointer with the magnet on the								Mean θ	Tan θ	M Weber/ cm	H Oersted
		East arm				West arm							
		North Pole towards needle		South Pole towards needle		North Pole towards needle		South Pole towards needle					
		θ_1	θ_2	θ_3	θ_4	θ_5	θ_6	θ_7	θ_8				

Table – II

Determination of time period (T)

S.No.	No. of Oscillations "n"	Time taken (t) Sec.	Time period " T " = t/n sec.
Mean			

RESULT:

PART-I: The value of magnetic moment of the given magnet = ___________________weber/cm.

PART-II: The value of the horizontal component of earth’s magnetic field in the laboratory at Gcet-Keesara =_______________Oersted.

VIVA QUESTIONS:

1. What is a magnetic moment?

2. Define the horizontal component of Earth's magnetic field?

3. What is a deflection magnetometer used for?

4. Why do we use a bar magnet in this experiment?

5. What is the unit of magnetic moment?

6. Why is it important to avoid nearby magnetic materials during the experiment?

Experiment No: **Date:**

8. (a) V-I and L-I characteristics of LASER Diode

Aim: To Study and Plot the V-I and L-I Characteristics of the Laser Diode.

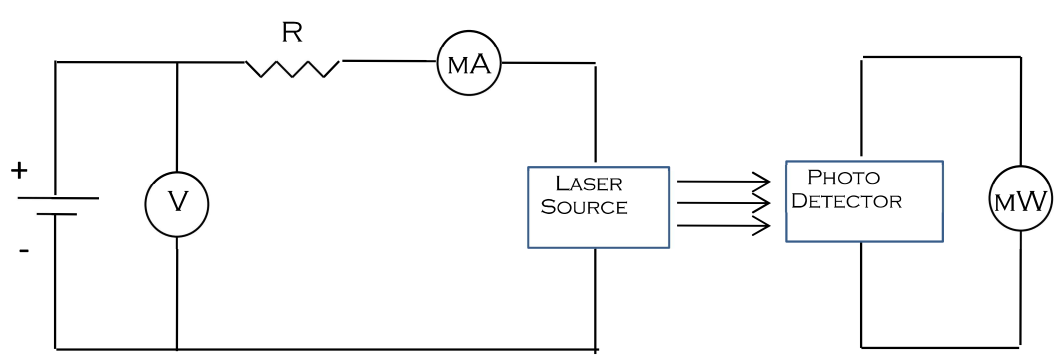
Apparatus: Laser diode, DC power supply, voltmeter, milliampere meter, milli Wattmeter (0-5mW)

Description:

Laser devices produce intense beams of light which are monochromatic, coherent, and highly collimated. The wavelength (color) of laser light is extremely pure (monochromatic) when compared to other sources of light, and all of the photons (energy) that make up the laser beam have a fixed phase relationship (coherence) with respect to one another. Light from a laser typically has very low divergence. It can travel over great distances or can be focused to a very small spot with a brightness that exceeds that of the sun. Because of these properties, lasers are used in a wide variety of applications in all walks of life.

Laser is a semiconductor diode made by creating of a junction of n-type and p-type materials. Thus, the principle of a Laser Diode action works precisely the same way that we described the creation of permanent light radiation: the forward-biasing voltage, V, causes electrons and holes to enter the depletion region and recombine. Alternatively, we can say that the external energy provided by V excites electrons at the conduction band. From there, they fall to the valence band and recombine with holes. Whatever point of view you prefer, the net result is light radiation by a semiconductor diode

Circuit Diagram:



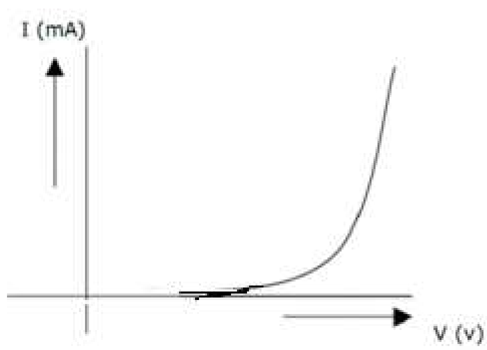
Procedure:

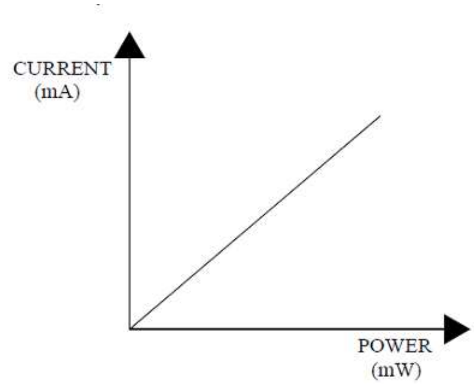
1.Connect the circuit as per the above diagram
2.Slowly increase supply voltage using a variable power supply
3.Now note down the Voltage, Current & Wattmeter values in a tabular form
4.Plot the graphs Voltage vs Current & Current vs Wattmeter
5.Do not connect the laser diode directly to the power supply as it may get damaged.

Observations:

S.No	Voltage (Volts)	Current (mA)	Power (mW)

Model Graph:





Result: Study and plot the V-I and L-I Characteristics of the laser diode.

(b) Wavelength of a LASER - Using diffraction grating

Aim: To determine the wavelength of a given source of laser using a plane transmission grating by the normal incidence method.

Apparatus: Plane diffraction grating, laser source, scale and grating mount.

Formula: $\lambda = \frac{2.54\ sin\theta}{nN} A^0$.

Where - λ is wavelength of given laser source.

N – Number of lines per inch n – Order of diffraction

θ- angle of diffraction.

Theory: A plane diffraction grating is a glass plate with equidistant fine parallel lines drawn very closely on one side of it. The number of lines drawn per inch is displayed on the diffraction grating by the manufacturers and is called LPI.

Thus, a grating consists of a large number of parallel slits of the same width (e) and separated by equal opaque space (d), and (e+d) is known as the grating element. If the grating has N rulings per inch, there will be N interspaces.

Hence,

Therefore N (e+d) = 1”= 2.54 cm, i.e. the grating element (e+d) = $2.54/N$ cm

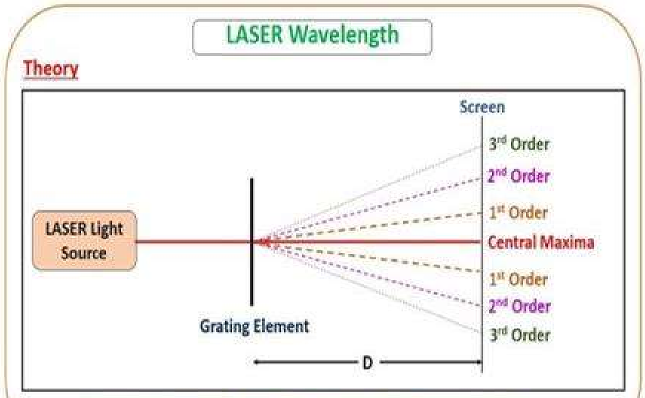
In normal incidence, the condition for obtaining principle maxima is (e+d) sinθ= n λ

or it can be shown that $\lambda = \frac{\mathbf{2.54\ sin\theta}}{\mathbf{nN}} A^0$.

Procedure:

Keep the grating in front of the laser beam such that light is incident normally on it. When laser beam is incident on the grating a diffraction pattern can be seen on the screen. The diffraction pattern consists of a central bright spot and several spots of varying intensity on either side of the central bright spot. The spot next to central maxima is called the first order maxima and the spot next to first order spot is second order maxima.

Measure the distance between the grating and the screen as 'D' and the distance between central maxima to first order diffraction pattern on either side of the central maxima are noted as d_1 and d_2. Similarly, the distance of second order diffraction spots is also noted and the readings are tabulated. From the readings, λ of the laser source can be determined.



Schematic Diagram of LASER diffraction by plane diffraction grating

Observations:

Number of lines on the grating, N = _________

The distance between the grating and the screen, D = ________________cm

S.No.	*Order*	*LHS* d_1 *cm*	*RHS* d_2 *cm*	*Mean* $d = \frac{d_1+d_2}{2}$ cm	$Sin\theta = \frac{d}{\sqrt{d^2+D^2}}$	$\lambda = \frac{2.54\, sin\theta}{nN}$ Å
1						
2						
3						
4						
5						
6						

Average λ = Å

Precautions:

1. Do not look at the laser beam directly.
2. The grating should be set normal to the incident laser beam.

Applications:

1. In material processing such as cutting, drilling and welding.
2. For pollution monitoring and remote sensing.
3. In holography.
4. In compact discs for storing and retrieving data.
5. In medicine for retina attaching, eye lens curvature correction and bloodless surgery.

Result: The wavelength of a given LASER beam is _______________________ A^0.

VIVA VOCE

1. **What is Coherence?**
2. **What is meant by Monochromatic light?**
3. **Define Intensity.**
4. **Define Life time.**
5. **What does 'LASER' stand for?**
6. **What is Stimulated Emission?**
7. **What is Spontaneous Emission?**
8. **What is a Metastable state?**
9. **Define Population Inversion.**
10. **Explain the Active medium.**
11. **Define Diffraction.**

Experiment No: **Date:**

9. B-H curve of a ferromagnetic material

Aim: To plot the B–H curve of a given ferromagnetic specimen by varying the magnetizing current and recording the corresponding magnetometer deflections.

Apparatus: Two Solenoid coils S & C, ferromagnetic specimen (soft iron rod), compensate coil, Constant current DC power supply, Rheostat (for varying current), Ammeter (0-3 Amperes), commutator, Deflection Magnetometer, Connecting wires.

Formula:

A ferromagnetic specimen placed in a solenoid is magnetized by the magnetic field produced by the current through the coil. The magnetizing field **H** inside the solenoid is given by: H= (Ni)/l

where:

N = number of turns in the solenoid
i = current through the coil (A)
l = length of the specimen (m)

The magnetic induction **B** is proportional to the magnetic moment of the specimen and is determined from the magnetometer deflection θ using tangent law.

$$B = B_H \tan \theta$$

B_H is the horizontal component of the earth's magnetic field, taken from the standard tables. By varying the current and noting the deflection, a plot of **B** vs **H** (B–H curve) can be drawn, which shows how the material responds to magnetization and helps in studying hysteresis effects**.**

Circuit diagram:

Where C: compensating coil; N: Magnetometer; Rh: Rheostat; R: Reversing key; A: Ammeter

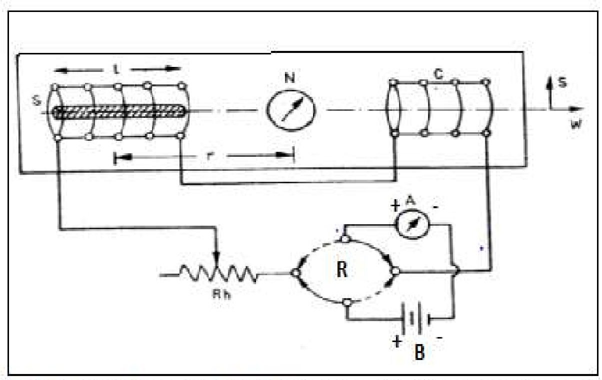


Fig. 1(a) Circuit diagram to draw B-H curve of the ferromagnetic material

Procedure:-

1. Connect the circuit as shown in the experimental setup diagram (solenoid in series with ammeter, rheostat, and DC power supply; magnetometer placed near the specimen).
2. Insert the ferromagnetic specimen in the solenoid.
3. Align the magnetometer so that its needle is parallel to the magnetic meridian (zero position).
4. Pass a small current through the solenoid and note the magnetometer deflection θ.
5. Increase the current in suitable steps using the rheostat and record the corresponding magnetometer deflections.

6. After reaching the maximum current, reduce the current in steps and again note the deflections to study hysteresis.
7. Repeat the steps 5 & 6 by reversing the direction of current in the circuit using commutator.
8. Calculate **H** from the solenoid parameters and current, and **B** from magnetometer readings using proportionality method (Tangent law).
9. Plot **B** (y-axis) against **H** (x-axis) to obtain the B–H curve.

Theory:

The phenomenon by which the magnetic induction (B) lags behind the magnetizing field (H) is called hysteresis. The hysteresis curve with H on x axis and corresponding B (of the material) on y- axis is shown in Fig-1(b).

Ferromagnetic materials like iron, cobalt and nickel are having certain specific magnetic properties. They will be having a large number of small regions in which magnetization will be already saturated spontaneously. These regions are called "Domains". Different domains will be having their magnetization in different directions and the overall magnetization with no applied magnetic field will be zero.

When an external magnetic field H is applied, the domains having their magnetization parallel to the field grow at the expense of other domains. Thus, initially the magnetization increases. This is shown by the curve OP. As and when all the domains get magnetized, there is no further scope for increase in magnetization and the magnetization is saturated as can be seen from PP^1.

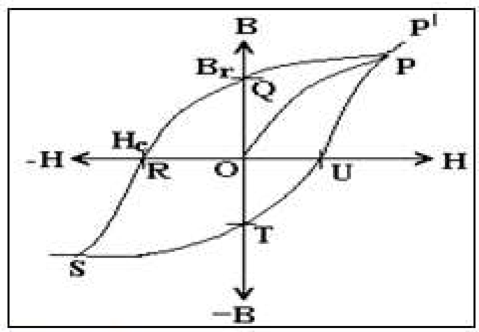


Fig. 1(b). B-H loop of a ferromagnetic material

Now let us gradually decrease the magnetizing field H. instead of coming back along PO, the curve retraces along PQ. At Q, even though the applied field H is completely removed, there still remains certain magnetic induction B_r (=OQ). This is because, the domain boundaries do not move completely back to their original position. This B_r is called the retentivity of the material.

Now, when the applied field is reversed in direction (-H), the remanent magnetization decreases and becomes zero at –H=OR. OR=H_c is called the corecivity. As we further increase H in reverse direction, the specimen gets magnetized now in an opposite direction and this process continues up to saturation at S. when the negative field is gradually decreased the curve follows the path ST where there is still magnetization (now is negative direction) even when H= 0.B becomes zero when +H is increased up to U. With further increase of +H the curve follows UP. The closed curve PQRSTUP is called the hysteresis curve of the given ferromagnetic material. The phase lag between B and H is the cause of hysteresis loop.

The tendency of the magnetic domains to turn around (both during magnetization and demagnetization as well) gives rise to mechanical stresses inside the material. This gives rise to heating. The energy wasted due to the cyclic magnetization is called the hysteresis loss and is given by the area under the hysteresis curve. When we apply an alternating current, the material gets magnetized during one half cycles and demagnetized during the second half cycle. Thus we get the hysteresis curve in a simple fashion.

Observations

1. Distance, r = ____________________________ m
2. Length of specimen, l = ___________________ m
3. No. of turns of solenoid, N=   turns
4. Horizontal component of earth's magnetic field, $B_H = 0.37 \times 10^{-4}$ Tesla

Observation Table 1

S.No	Current I (A)	Magnetic Field H= (Ni)/l	Deflections while increasing the current		Mean θ	Tan θ	Magnetic Induction B = B_H Tan θ	Deflections while decreasing the current		Mean θ	Tan θ	Magnetic Induction B = B_H Tan θ
			θ_1	$\theta 2$				$\theta 3$	$\theta 4$			
1	0.25											
2	0.5											
3	0.75											
4	1											
5	1.25											
6	1.5											
7	1.75											
8	2											
9	2.25											
10	2.5											

Observation Table 2 (After reversing the direction of current)

S.No	Current i (A)	Magnetic Field H= (Ni)/l	Deflections while increasing the current		Mean θ_R	Tan θ_R	Magnetic Induction B = B_H Tan θ_R	Deflections while decreasing the current		Mean θ_R	Tan θ_R	Magnetic Induction B = B_H Tan θ_R
			θ_1	$\theta 2$				$\theta 3$	$\theta 4$			
1	0.25											
2	0.5											
3	0.75											
4	1											
5	1.25											
6	1.5											
7	1.75											
8	2											
9	2.25											
10	2.5											

Graph

To Plot the B–H curve, take experimental readings of magnetic field strength H (A/m) along the horizontal axis (x-axis) and magnetic flux density B (Tesla) along the vertical axis (y-axis), then join the points in sequence for increasing and decreasing H to obtain the hysteresis loop.

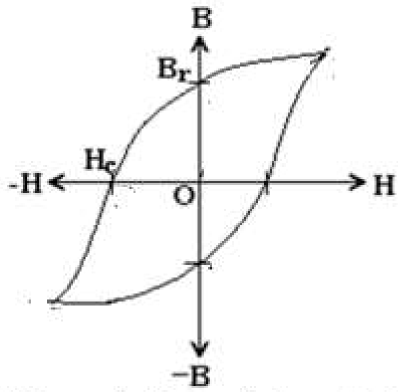


Fig. 1(c). Magnetic Hysteresis Loop (B–H Curve)

Precautions

- Ensure the magnetometer is properly levelled and aligned with the magnetic meridian.
- Increase and decrease the current gradually to avoid jerks.
- Avoid tapping or vibrating the setup during readings.
- Calibrate the magnetometer before starting measurements.

Result

- The B–H curve of the given ferromagnetic material was obtained.
- 1) B_r = retentivity = ______________Tesla (Wb/m^2)
- 2) H_c = corecivity = _______________Ampere/meter.
- 3) Hysteresis loss = _________________________joules/cycle/unit volume.

Viva voce

1. What is hysteresis?
2. In what materials do you find this behaviour?
3. What is the root cause for hysteresis?
4. What are magnetic domains?
5. How is the curie law modified due to these domains?
6. What happens to a ferromagnetic material above the curie temperature θ_C.
7. What is the cause of hysteresis?
8. What is the cause of hysteresis loss?
9. How can you estimate the hysteresis loss of a given specimen?
10. What are the practical uses of hysteresis curves?
11. What are the units of B and H?

10. Study of the P-E loop of a given ferroelectric crystal.

Aim: To study the ferroelectric hysteresis behavior of Given ferroelectric material (TiO_3) and determine Saturation Polarization (Ps), Remanent Polarization (Pr), Coercive Field (Ec), and Energy Loss per cycle.

Apparatus:

1. P-E Hysteresis setup with power supply
2. Barium Titanate sample (Thickness d=1.8mm, Diameter=10.5mm)
3. Cathode Ray Oscilloscope (CRO)
4. Connecting wires and probes
5. Sample holder with spring-loaded probe
6. Ruler for screen measurements

Theory:

Ferroelectric Hysteresis Concept

Ferroelectric materials exhibit a lag between applied electric field (E) and resulting polarization (P), creating a hysteresis loop. This memory effect is crucial for applications in memory devices and capacitors.

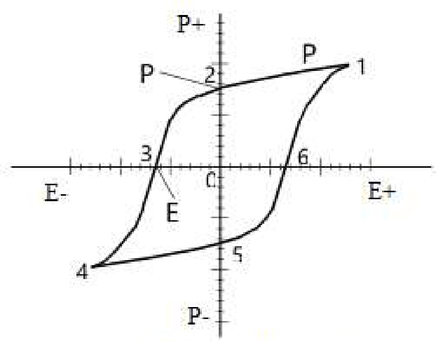


Fig 1 : The P - E curve obtained by cyclic polarization of ferroelectric ceramic.

Key Parameters from Hysteresis Loop:

- **Ps**: Maximum polarization when all dipoles are aligned
- **Pr**: Residual polarization when electric field is zero
- **Ec**: Electric field required to reduce polarization to zero
- **Hysteresis Loss**: Energy dissipated per cycle = Area inside loop

Formulae:

1. ***Electric Field Calculation:***

$$E = \frac{V}{d} \text{ (V/m)}$$

Where: d=1mm=0.001m

*2. **Polarization Calculation:***

$$P = \frac{Q}{A} \quad (C/m^2)$$

Where electrode area $A = \pi r^2 = \pi(5.25\times10^{-3})^2 = 8.66\times10^{-5} m^2$

*3. **Energy Loss Calculation:***

$$E_{loss} = \frac{0.5 x S_v S_h x\ Area}{R x d} \ (J/s/m^2)$$

Where:

- Sv = Vertical sensitivity (V/div)
- Sh = Horizontal sensitivity (V/div)
- Area = Loop area in mm^2
- R = Variable resistor value (Ω)
- d = Sample thickness (m)

Circuit diagram:

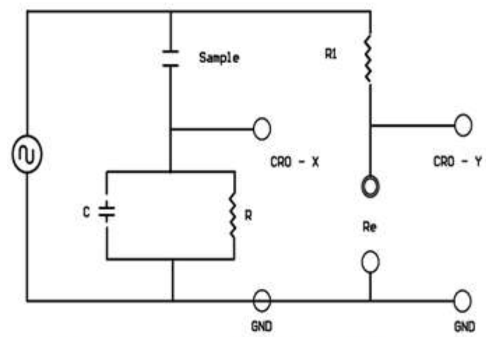


Figure: Circuit diagram to obtain P-E loop of ferroelectric material

Constants for this setup:

Constant	Value	Description
Horizontal Amplitude Factor	0.2	For X-axis corrections
Vertical Amplitude Factor	0.5	For Y-axis corrections
Sample Thickness (d)	1 mm	Fixed sample dimension
Sample Diameter	10.5 mm	Fixed sample dimension
Conversion Constant (Kv)	16.66×10^3	Polarization conversion

Experimental procedure:

A. INSTRUMENT CONNECTIONS

1. Connect **X-output → CRO X-input**
2. Connect **Y-output → CRO Y-input**
3. Set CRO to **X-Y mode**
4. Power ON all instruments

B. CALIBRATION PROCEDURE

Step 1: Horizontal Calibration

- Set X-input sensitivity: **50 mV/div**
- Observe horizontal line on CRO
- Measure length with ruler: **L_x = ____________________ cm**
- Calculate: Horizontal Raw Factor = **50 mV** ×L_x = ___________________ V-cm

Step 2: Vertical Calibration

- Engage X-ground at CRO input
- Set Y-input sensitivity: **2V/div**
- Measure length with ruler: **L_y = _____________________ cm**
- Calculate: Vertical Raw Factor = **2 V** ×L_y = _______________________ V-cm

Step 3: Calculate Final Sensitivities

S_h = (Horizontal Raw Factor ÷ 2) × 0.2 = _____________________ V/div

S_v = (Vertical Raw Factor ÷ 2) × 0.5 = ____________________________ V/div

C. HYSTERESIS LOOP MEASUREMENT

1. Disengage X-ground
2. Select low scan frequency
3. Adjust voltage control for ±4 swing
4. **Rotate resistor knob** to obtain perfect loop shape
5. Center loop using X-Y position controls

MEASUREMENTS AND CALCULATIONS:

A. OBSERVATION TABLE

Parameter	Symbol	CRO Reading (cm)	Calculated Value	Units
Saturation Polarization	Ps			C/m^2
Remanent Polarization	Pr			C/m^2
Positive Coercive Field	+Ec			V/m
Negative Coercive Field	-Ec			V/m

B. MEASUREMENT FROM LOOP

Measure on CRO screen:

- **Ps** = Vertical distance from center to top = ______________ cm
- **Pr** = Vertical distance at E=0 = _______________ cm
- **+Ec** = Horizontal distance from center to right crossing = __________________ cm
- **-Ec** = Horizontal distance from center to left crossing = __________________ cm

C. CALCULATION EXAMPLES

Example Data:

L_x = ______________ cm, L_y = _______________ cm

P_s = ____________ cm, E_c = _________________ cm

1. Calculate Sensitivities:

S_h = (50mV/div × _________________ ÷ 2) × 0.2 = ___________________ V/div

Sv = (2V/div x ___________________ ÷ 2) × 0.5 = __________________ = 1.55 V/div

2. Calculate Electric Field (Ec):

E_c = (E_c in cm × S_h) / d

= (__________________ × _________________) / 0.001

= ____________________ / 0.001 = ___________________ V/m

3. Calculate Polarization (Ps):

$Ps = (Ps \text{ in cm} \times S_v) \times 16{,}660$

$= (\_\_\_\_\_\_\_\_\_\_\_\_ \times \_\_\_\_\_\_\_\_\_\_\_\_) \times 16{,}660$

$= \_\_\_\_\_\_\_\_\_\_ \times 16{,}660 = 103{,}292 \text{ C/m}^2$

D. ENERGY LOSS CALCULATION

Step 1: Measure Loop Area

- Count small squares inside hysteresis loop
- 1 small square = 1 mm^2
- Total Area (A) = ____________ mm^2

Step 2: Calculate Energy Loss

$E_{loss} = 0.5 \times S_v \times S_h \times A \times 10^{-6} / (R \times d)$

S_v = ____________ V/div, S_h = ____________ V/div

A = ____________ mm^2, R = 250 Ω, d = 0.001 m

$E_{loss} = 0.5 \times \_\_\_\_\_\_\_\_ \times \_\_\_\_\_\_\_\_ \times \_\_\_\_\_\_\_\_ \times 10^{-6} / (250 \times 0.001)$

$= \_\_\_\_\_\_\_\_ / \_\_\_\_\_\_\_\_ = \_\_\_\_\_\_\_\_ \text{ J/s/m}^2$

PRECAUTIONS

1. **Sample Handling**: $BaTiO_3$ is fragile - handle gently
2. **Electrical Safety**: Do not touch holder during operation - risk of shock
3. **Connections**: Ensure secure connections to avoid noise
4. **Calibration**: Perform calibration carefully for accurate results
5. **Loop Adjustment**: Rotate resistor knob slowly for optimal loop shape
6. **Warm-up Time**: Allow 10-15 minutes for instrument stabilizations

RESULT

CALIBRATION DATA:

Lx = ____________________ cm

Ly = ____________________ cm

Vy = ___________________ Vpp

S_h = ____________________ V/div

S_v = ____________________ V/div

LOOP MEASUREMENTS:

Ps = ________________ cm → ____________________ C/m²

Pr = ________________ cm → ____________________ C/m²

+Ec = ___________ ___ cm → ____________________ V/m

-Ec = _______________ cm → ____________________ V/m

ENERGY LOSS:

Loop Area = __________________ mm²

Resistor R = ___________________ Ω

Energy Loss = _________________ J/s/m²

APPLICATIONS

- **Non-volatile Memory** (FeRAM)
- **High-density Capacitors**
- **Piezoelectric Sensors**
- **Microwave Devices**
- **Pyroelectric Detectors**

Viva voce

1. What is ferroelectric hysteresis and how does it differ from magnetic hysteresis?

2. Explain the physical significance of Pr, Ps, and Ec.

3. Why does the hysteresis loop have area? What does this area represent?

4. What are the practical applications of ferroelectric materials?

5. Why is calibration important in this experiment?